

CIDEX: The Compression-Ignition Engine Certification Panel — Harmonization Pipeline

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Abstract

The United States Environmental Protection Agency publishes emission certification records for compression-ignition engines as two Microsoft Excel workbooks whose internal structures are mutually incompatible. One is organised long on pollutant and wide on test type; the other is wide on pollutant beneath a two-level header, with family attributes distributed on both sides of the measurement block. Both place their true column headers on the second row, so the naive read of either file produces plausible but incorrect data without raising an error.

This report documents CIDEX, a pipeline that reconciles both sources into a single analysis-ready panel of **8,627 engine families**, **9,794 engine configurations** and **94,869 emission records** across **108 manufacturers** and model years 2011–2027. It also documents a resolved carryover lineage linking **2,541 distinct certification families** across model years, a structure present in the nonroad source but not, as far as the author is aware, previously published in resolved form.

The report states the method in enough detail to be re-implemented independently, records every design decision with its justification, and — in Section 6 — documents the defects found in the pipeline during development and how each was detected. That section is included because a validation record that reports no defects is not evidence of a correct pipeline; it is evidence of an incurious one.

Document. Technical report, part of the [FACET](https://github.com/osariemenimafidon/cidex) research program. Repository <https://github.com/osariemenimafidon/cidex>. Licence: data and documentation CC BY 4.0; code MIT. Every figure in this report is interpolated from the pipeline’s own `stats.json`.

1. Introduction

1.1 Motivation

Certification data is the only comprehensive public record of what engines were approved to emit, by whom, and under which standard. It underpins questions about fleet composition, technology adoption, regulatory stringency over time, and the relationship between certified and in-service behaviour.

In practice the data is difficult to use. The obstacle is not access — EPA publishes the files openly and without authentication — but structure. Every researcher who wants to ask a cross-cutting question must independently reverse-engineer two different layouts, reconcile two pollutant encodings, and work out that the file’s headers are not where a spreadsheet reader will look for them. That work is duplicated, undocumented, and performed under no obligation to get it right.

CIDEX performs that reconciliation once, publicly, with provenance.

1.2 Scope

CIDEX v1.0.0 covers the two current EPA certification files: heavy-duty highway gasoline and diesel engines (model years 2015 onward) and nonroad compression-ignition engines (model years 2011 onward). EPA’s archive files — highway 1982–2016 and nonroad 1996–2011 — are **not** ingested in this version; their omission has a measurable effect on coverage which is quantified in Section 7.3 and carried as a flag in the data.

1.3 What this report is for

It should be possible to re-implement CIDEX from this report alone and obtain the same numbers. Where a choice was made that a competent person could reasonably have made differently, the choice is stated as a choice rather than presented as the only option.

2. Sources

2.1 The two workbooks

	Highway	Nonroad
Coverage	Heavy-duty highway gasoline and diesel	Nonroad compression-ignition
Model years	2015–2026	2011–2027
Source rows (Family Info)	7,804	10,214
Unique families	702	7,925
Manufacturers	27	86
Orientation	Long on pollutant, wide on test type	Wide on pollutant, two-level header

Both are US Government works and carry no copyright restriction on reuse.

2.2 Access constraint and the provenance record

The build environment used for this work cannot reach `epa.gov`: outbound requests are refused by an organisational egress policy. The source files were therefore downloaded manually and placed in `data/raw/`.

This is a weakness — an automated fetch is verifiable in a way a manual one is not — and it is mitigated rather than hidden. `scripts/01_provenance.py` records, for every raw file, its name, byte count, SHA-256 digest, modification time, and the canonical URL it is asserted to have come from. A rebuilder does not have to accept the assertion: they download the file themselves and compare digests. The full record is Appendix A.

3. The harmonization problem

3.1 Headers are on the second row

Both workbooks reserve row 1 for a merged group header and place actual column names on row 2. `pandas.read_excel` defaults to `header=0`. Applied to either file it returns a frame whose column names are `Unnamed: 0 ... Unnamed: 44` and whose first data row contains the real headers. Nothing raises. Any subsequent selection by column name fails or, worse, succeeds against the wrong column.

3.2 The two files are organised in opposite directions

The **highway** file is long on pollutant: one row per family, engine configuration and pollutant, with a `Pollutant Name` column taking 10 distinct values. It is simultaneously *wide* on test type — transient and steady-state results occupy separate column blocks on the same row.

The **nonroad** file is the reverse. One row carries all pollutants as columns, arranged in four groups beneath a two-level header: `steady_state`, `transient`, `fel`, `smoke`. Family attributes appear both *before* and *after* the measurement block, which defeats any approach based on column position alone.

Reconciling these into one schema is the substance of the pipeline.

3.3 The engine family name is a true key

In both panels the count of distinct engine family names equals exactly the count of distinct (model year, family) pairs — 702 and 7,925 respectively. The model year is encoded in the family name's first character, so the name alone identifies the family-year. No surrogate key is required, and a name collision across years is impossible by construction.

3.4 Family attributes and configuration attributes are different things

A single engine family may be certified across several engine configurations. In the nonroad source, 1,487 of 7,925 families occupy more than one row — up to 34 for a single family — differing by engine code, model, displacement or test procedure.

Placing those attributes in a family-level table would force an arbitrary row to win and silently discard the rest. CIDEX separates them: attributes constant within a family go to the family table; attributes that vary go to a configuration table. The split is not asserted — the build **tests** it and aborts if any attribute in the family table varies within a family. Before the split, 7 nonroad attributes varied; after it, 0 do.

4. Method

4.1 Pipeline overview

Script	Function
01_provenance.py	Hash and log every raw file before use
02_harmonize_highway.py	Highway source → CIDEX schema
03_harmonize_nonroad.py	Nonroad source → CIDEX schema
04_carryover.py	Resolve nonroad certification lineage
05_combine_qa.py	Combine, flag, check, write <code>stats.json</code>
06_docs.py	Regenerate the documentation set
07_publish_gate.py	Pre-publication scan
08_figures.py	Figures
09_release.py	Build the deposit archive

Script	Function
<code>11_report.py</code>	Generate this report

Scripts are numbered in run order and each is independently re-runnable.

4.2 Highway harmonization

The highway source is read with `header=1` — the second row. Family-level attributes are projected into a family table; configuration attributes (engine code, test model, engine identifier, test fuel, test date) into a configuration table.

The pollutant name is mapped through a controlled vocabulary. If any source value fails to map, the script **raises** rather than dropping the row: a silently discarded pollutant is exactly the failure mode this pipeline exists to prevent. In this build, 0 values failed to map.

The test-type block is then unpivoted. Each source row yields one record per test type, and records with no value in any measurement column are removed as unpivot artefacts rather than facts: 15,608 rows before, 2,184 dropped, 13,424 retained.

Family-attribute constancy is tested, not assumed. In this build 0 highway family attributes varied within a family.

4.3 Nonroad harmonization

The nonroad two-level header is resolved without hard-coding column positions. The group row is forward-filled, then a column is treated as a measurement **only if** its row-2 sub-name is one the group actually contains. This is what keeps **Engine Model** — which sits immediately after the FEL block and inherits its forward-filled group label — classified as an attribute rather than as a pollutant. The classifier found 22 measurement columns and 24 attribute columns, with 0 attributes unmapped.

Wide-to-long expansion produces 224,708 candidate records, of which 141,657 are empty cells in a sparse layout and are dropped, leaving 81,445.

Family Emission Limits are a limit, not a measured result, and are moved to their own column. **The merge is keyed on source-row identity, not on the family key** — see §6.1.

4.4 Carryover lineage

5,982 nonroad families name a predecessor family. These references chain, so a family’s certification history can be traced across model years.

Three cases are handled explicitly rather than by exception:

- **Self-reference.** 30 families name *themselves* as their own carryover. EPA appears to use this to mark an unchanged re-certification. A naive traversal reads it as a cycle; CIDEX treats it as a lineage terminus.
- **Orphans.** 568 families name a predecessor absent from this panel, usually pre-2011 and outside the published window. The lineage is truncated, not wrong — so a depth of 0 does not always mean “first of its line”.
- **True cycles.** 0 detected. The graph is a forest.

Result: 2,541 distinct lineages, mean depth 1.382, maximum 14 model years. The deepest runs CCPXL18.1NYS → TCPXL18.1NYS (MY2026).

4.5 Units: no silent conversion

The two panels certify in different units. Highway results are in **g/bhp-hr**, nonroad in **g/kW-hr**, and nonroad smoke opacity in **percent**. Every record carries its units in a column; **no conversion is performed**.

This is deliberate. A dataset that silently normalised units would let a user aggregate across panels and obtain a plausible, wrong number with no indication anything was amiss. Conversion is available but must be invoked:

```
from cidex.units import to_g_per_kWh
to_g_per_kWh(0.20, "g/bhp-hr") # 0.2682
```

`cidex.units.headroom()` raises `MixedUnitsError` rather than computing across unit systems. The constant used is $1 \text{ hp} = 0.7456998716 \text{ kW}$.

4.6 Quality flags and coverage flags

Two flags are carried on the data rather than left to documentation:

`quality_flag` marks the 50 records (0.0527% of the panel, across 18 families, all nonroad) whose certification result is **negative**. An emission rate cannot be physically negative. CIDEX does not correct, clip or drop them — the published value stands as EPA reports it and carries a flag. Whether this reflects a data-entry issue or an undocumented encoding is unresolved.

`year_coverage` marks model years that are structurally incomplete (§7.3).

5. Validation

5.1 Integrity checks

10 checks run on every build. All currently pass.

Check	Result	What it tests
<code>family_key_unique</code>	PASS	No (panel, model year, engine family) triple appears twice in the family table.
<code>no_emission_row_without_family</code>	PASS	Every emission record joins a family row; no orphan measurements.
<code>units_never_null</code>	PASS	The units column is populated on every emission record.
<code>no_mixed_units_within_panel</code>	PASS	No panel carries more than its expected unit systems.
<code>highway_units_evidenced_by_standards</code>	PASS	The modal certification standard per pollutant in the highway panel matches the known US heavy-duty standards expressed in g/bhp-hr. See §5.2.
<code>highway_units_labelled</code>	PASS	Every highway record is labelled g/bhp-hr.
<code>nonroad_units_correct</code>	PASS	Nonroad records carry only g/kW-hr or pct opacity.
<code>carryover_no_true_cycles</code>	PASS	The carryover lineage graph contains no cycle.
<code>every_negative_value_is_flagged</code>	PASS	Every negative certification result carries a quality flag.

Check	Result	What it tests
<code>flagged_share_below_0pt1pct</code>	PASS	Flagged records remain under 0.1% of the panel; a sharp rise would indicate a parsing regression.

These test **internal consistency**. They cannot establish that a value matches EPA’s record; that is what §5.3 is for.

5.2 The units evidence test

The highway source file states its units **nowhere**. Its headers are bare — **TR Cert Result, Transient Standard**. The claim that highway results are in g/bhp-hr was originally an assertion in the documentation.

It is now a test. US heavy-duty highway standards have known values in g/bhp-hr, and the modal certification standard per pollutant in the panel is compared against them:

Pollutant	Known standard (g/bhp-hr)	Modal value in CIDEX
NOx	0.2	0.2
PM	0.01	0.01
NMHC	0.14	0.14
CO	15.5	15.5

Expressed in g/kW-hr the same standards would be 0.268, 0.013, 0.188, 20.786 — an order of magnitude away in every case, so the test cannot pass by coincidence. It runs on every build.

5.3 The author verification gate

Mechanical checks cannot establish correspondence with the source. That requires a person to look up specific records in EPA’s own certificate tool and compare. The pipeline therefore ships five named spot-checks, each probing a different failure mode:

#	Panel	Family	MY	Pollutant	CIDEX value	What it probes
1	highway	MSZXH05.23FD	2021	NOx	0.17 g/bhp-hr	highway, transient NOx — the standard case
2	highway	FNGCH0466AEA	2015	PM	0.01 g/bhp-hr	highway, earliest model year present — tests the MY2015-16 archive boundary

#	Panel	Family	MY	Pollutant	CIDEX value	What it probes
3	nonroad	LCEXL60.0AAB	2020	NOx	0.48 g/kW-hr	nonroad, steady-state NOx — tests the wide-to-long unpivot deepest carryover lineage (depth 14, root CCPXL32.0NZS) — tests lineage resolution
4	nonroad	TCPXL32.1NZS	2026	NMHC	0.08 g/kW-hr	a flagged negative value — is this real in EPA's own record?
5	nonroad	CDICL05.8HTA	2012	NOx	-1.0 g/kW-hr	

Publication is blocked until the author signs `docs/VERIFICATION_CHECKLIST.md`. `scripts/09_release.py` refuses to build a deposit archive while the attestation file is absent, and every document and figure regenerates stamped DRAFT without it.

Status: Signed by Imafidon, Osariemen (ORCID 0009-0006-3069-4674) on 2026-09-15, recorded by the presence of `.gate-signed` in the repository root.

6. Defects found and corrected

Three defects were found during development. They are recorded because a validation section reporting none would say more about the thoroughness of the review than about the correctness of the pipeline.

6.1 The FEL merge multiplied rows

Severity: data corruption. Detected: adversarial review before publication.

Family Emission Limits were merged onto the emission records using `(model_year, engine_family, pollutant)`. Because 1,487 families occupy more than one source row, that key matched several FEL entries per family and the merge fanned every measurement row out to match.

The result was **2,139 phantom rows** — genuine values, duplicated. This is worse than missing data: no nulls appeared, no outliers, nothing downstream looked wrong. It inflated every count by roughly 2.6%.

Corrected by carrying source-row identity through the wide-to-long expansion and merging on it. The script now aborts if the row count changes across the merge. Nonroad emission records fell from 83,584 to 81,445.

6.2 The units claim was asserted, not evidenced

Severity: unverified claim in published documentation. Detected: same review.

g/bhp-hr appeared in the codebook as fact, having been inferred rather than read. The inference was correct, but the reasoning was not sound — the same reasoning, applied to a sibling project, substituted a European fuel standard for a US regulatory one and produced a wrong result. Corrected as described in §5.2.

6.3 Carryover self-references counted as cycles

Severity: incorrect summary statistic. Detected: inspecting the output.

The first lineage traversal reported 109 cycles. Every one was a family naming itself — a terminus, not a loop. Corrected by treating self-reference as a stopping condition and detecting true cycles with a visited set. True cycles: 0.

7. Results

7.1 Panel contents

	Rows
<code>cidex_family.csv</code>	8,627
<code>cidex_config.csv</code>	9,794
<code>cidex_emissions.csv</code>	94,869
<code>cidex_carryover.csv</code>	7,925

108 manufacturers · 14 pollutants · 3 test types · units present: g/bhp-hr, g/kW-hr, pct opacity

7.2 Pollutant coverage

Pollutant	Records
CH4	7,456
CO	14,238
CO2	13,196
HCHO	503
N2O	4,700
NH3	3
NMHC	14,219
NMHC_E	13
NMHC_NOx	6,712
NOx	14,238
PM	14,190
smoke_accel	1,853
smoke_lug	1,763
smoke_peak	1,785

NMHC_E (Non-Methane Hydrocarbon Equivalent) is kept distinct from NMHC. It is a different regulatory construct, and folding it in would destroy information that cannot be reconstructed downstream, where keeping it separate costs a user one filter.

7.3 Model-year coverage

Both panels are structurally incomplete at their edges, for two unrelated reasons. A year is flagged when it holds under 50% of the median count of that panel’s interior years.

Panel	Model year	Families	Share of median	Reason
highway	2015	3	5%	EPA archive not ingested
highway	2016	29	45%	EPA archive not ingested
nonroad	2011	66	12%	EPA archive not ingested
nonroad	2027	8	2%	still being certified

Usable ranges for trend analysis: highway **MY2017–2026**, nonroad **MY2012–2026**.

The threshold is a judgement. It separates the observed cases cleanly — no year sits near the boundary — but the raw counts are published so a reuser can choose differently.

8. Limitations

Stated in full in `docs/LIMITATIONS.md`. In summary: units are not comparable across panels without conversion; 50 records carry physically impossible negative values, published as EPA reports them; nonroad records have no pollutant-specific standard because the source states it only at family level; edge model years are incomplete; manufacturer names are not normalised; and carryover lineage is nonroad-only with 568 open edges.

9. Reproducibility

```
pip install -r requirements.txt
./verify.sh
```

`verify.sh` rebuilds from `data/raw/` and diffs every value against the shipped `docs/stats_reference.json`, excluding only the build date. A mismatch is a finding.

Every number in every CIDEX document — including this report — is interpolated from `data/processed/stats.json`, which the pipeline writes. The column mappings in Appendix B are imported from the pipeline modules at generation time, so they cannot drift from the code they document.

Appendix A — Provenance record

File	Bytes	SHA-256	Recorded
heavy-duty-gas-and-diesel-engines-2015-present.xlsx	32,008,765	8edfa15deef231828e2617302649fe58ade0e96eb8dfe044b32cc4e63	2026-03-15
nonroad-compression-ignition-2011-present.xlsx	13,996,420	9a78cf00ca4f930899fed3268109145f32dade525e8e68d5394b3df56	2026-03-15

Canonical sources:

- `heavy-duty-gas-and-diesel-engines-2015-present.xlsx` — <https://www.epa.gov/system/files/documents/2026-03/heavy-duty-gas-and-diesel-engines-2015-present.xlsx>
- `nonroad-compression-ignition-2011-present.xlsx` — <https://www.epa.gov/system/files/documents/2026-03/nonroad-compression-ignition-2011-present.xlsx>

Appendix B — Source-to-output column mapping

B.1 Highway family attributes

Source column	CIDEX column	Table
Manufacturer	manufacturer	family
Model Year	model_year	family
Engine Family	engine_family	family
Certificate No.	certificate_no	family
Date Issued	date_issued	family
Engine Cycle	engine_cycle	family
Fuel Type	fuel_type	family
Fuel Metering System	fuel_metering_system	family
Introduction Date	introduction_date	family
Useful Life	useful_life	family
Intended Service Class	intended_service_class	family

B.2 Highway measurement blocks

Source column	CIDEX column	Destination
TR Cert Result	cert_result	emissions, test_type = transient
Transient (TR) Comb Adj Result	adj_result	emissions, test_type = transient
Transient Standard	standard	emissions, test_type = transient
Transient FEL	fel	emissions, test_type = transient
Transient FCL	fcl	emissions, test_type = transient
SS Cert Result	cert_result	emissions, test_type = steady_state
Steady State (SS) Adj Result	adj_result	emissions, test_type = steady_state
SS Standard	standard	emissions, test_type = steady_state
SS FEL	fel	emissions, test_type = steady_state
SS FCL	fcl	emissions, test_type = steady_state

B.3 Nonroad attributes

Source column	CIDEX column	Table
Model Year	model_year	family or configuration
Engine Family	engine_family	family or configuration
Manufacturer	manufacturer	family or configuration
Certificate #	certificate_no	family or configuration
Issue Date	date_issued	family or configuration
Commerce Introduction Date	introduction_date	family or configuration
Carryover Engine Family Name	carryover_family	family or configuration
Power Category	power_category	family or configuration
Applicable Regulation	regulation	family or configuration

Source column	CIDEX column	Table
Applicable Tier	tier	family or configuration
Applicable Compliance Standard	compliance_standard	family or configuration
Fuel	fuel_type	family or configuration
Fuel Meter System	fuel_metering_system	family or configuration
Useful Life of Engine Family	useful_life	family or configuration
Engine Combustion Cycle	engine_cycle	family or configuration
Non Aftertreatment Device Type	non_aftertreatment_device	family or configuration
Aftertreatment Device Type	aftertreatment_device	family or configuration
Engine Model	engine_model	family or configuration
Engine Code	engine_code	family or configuration
Displacement	displacement_l	family or configuration
Certification Fuel	certification_fuel	family or configuration
Engine Operation	engine_operation	family or configuration
Test Procedure	test_procedure	family or configuration
Test Type	test_procedure_type	family or configuration

Appendix C — Pollutant vocabulary

Highway source values map as follows. Nonroad sub-headers map through a parallel table in `src/cidex/vocab.py`.

Source value	CIDEX code
Nitrogen Oxides	NOx
Carbon Monoxide	CO
Non-Methane Hydrocarbons	NMHC
Non-Methane Hydrocarbon Equivalent	NMHCE
Particulate Matter	PM
Carbon Dioxide	CO2
Methane	CH4
Nitrous Oxide	N2O
Formaldehyde	HCHO
Ammonia	NH3

Generated by `scripts/11_report.py` from `stats.json` on 2026-09-15.